

Suggested Project Topics

Students are required to complete a group (or individual, where approved) mathematical modelling project based on a real-world engineering or scientific problem. The following topics are suggested as suitable project areas. Students may also propose an alternative topic, subject to approval by the unit coordinator.

Fluid Mechanics

1. Water Tank Filling and Emptying
 - Develop a model for a storage tank with inflow and outflow.
 - Predict the variation of water volume with time.
2. Rainwater Harvesting System
 - Model rainfall, water usage and evaporation.
 - Investigate seasonal variations in water storage.
3. Pipe Flow
 - Develop a model for flow through a pipeline.
 - Investigate the effects of flow rate and pipe dimensions.

Heat Transfer

4. Cooling of Hot Drinks
 - Apply Newton's Law of Cooling.
 - Compare model predictions with experimental observations.
5. Heat Conduction in a Metal Rod
 - Solve the one-dimensional heat equation.
 - Investigate steady-state and transient temperature distributions.
6. Home Insulation
 - Model heat loss through walls.
 - Investigate the effect of insulation thickness.

Diffusion and Environmental Modelling

7. Pollution in a River
 - Develop a diffusion model for pollutant transport.
 - Predict downstream concentration profiles.
8. Drug Diffusion Through Skin
 - Model the diffusion of medication into biological tissue.
9. Salt Diffusion in Water
 - Investigate the diffusion of dissolved salt.

10. Lake Pollution

- Model contaminant concentration in a lake with inflow and outflow.

11. Groundwater Contamination

- Model contaminant transport through porous soil.

Population and Biological Systems

12. Population Growth

- Compare exponential and logistic growth models.

13. Predator–Prey System

- Investigate the Lotka–Volterra equations.

14. Disease Spread

- Develop and analyse a simple SIR epidemic model.

15. Fish Population Management

- Model harvesting strategies for sustainable fisheries.

Traffic and Transport

16. Traffic Flow on a Highway

- Develop a conservation law model for traffic density.

17. Traffic Signal Optimisation

- Model queue formation and waiting times at intersections.

18. Vehicle Braking

- Model stopping distance under different braking conditions.

19. Projectile Motion

- Compare ideal projectile motion with models including air resistance.

Industrial Applications

20. Mixing Tank

- Model the concentration of a dissolved substance in a continuously stirred tank.

21. Continuous Stirred Tank Reactor (CSTR)

- Develop a concentration model for a chemical reactor.

22. Water Filtration System

- Model contaminant removal in a filtration process.

23. Building Energy Consumption

- Model heating and cooling energy requirements.

24. Solar Water Heating System

- Develop a mathematical model for solar energy collection.